Heltec T114 v2

Overview

Heltec Mesh Node T114 (Rev. 2.0) is an IoT development board designed and produced by Heltec Automation(TM). It combines an nRF52840 microcontroller with an SX1262 LoRa transceiver, optional 1.14″ TFT display, and GNSS support. [redacted] See the Heltec Mesh Node T114 pages [1] for more details.

The features include the following:

- Microprocessor: Nordic nRF52840 (ARM Cortex-M4F, 64 MHz, 256 KB RAM, 1 MB flash)
- LoRa node chip: Semtech SX1262
- Optional onboard 1.14″ 135×240 TFT display (ST7789V, SPI)
- GNSS module interface (NMEA over UART)
- USB-C interface with USB device (CDC ACM) support
- BLE 5.0, IEEE 802.15.4 (OpenThread)
- One user button, one green LED, SK6812 RGB LED strip (2 LEDs)
- Li-Po battery & solar panel 2*1.25mm connector
- 2*13*2.54mm expansion pin headers

Board Overview



Heltec T114 v2

NAME	`heltec_t114_v2`
VENDOR	Chengdu Heltec Automation Technology Co., Ltd.
STATUS	Not actively maintained
ARCHITECTURE	arm
SOC	nrf52840

Browse board sources

Supported Features

The `heltec_t114_v2` board supports the hardware features listed below.

on-chip / on-board	Feature integrated in the SoC / present on the board.
×2 / ×2	Number of instances that are enabled / disabled. Click on the label to see the first instance of this feature in the board/SoC DTS files.

vnd,foo

Compatible string for the Devicetree binding matching the feature.
Click on the link to view the binding documentation.

heltec_t114_v2 / nrf52840

On-target memory for this board target: 256 KiB of RAM, 1 MiB of Flash.

TYPE	LOCATION	DESCRIPTION		COMPATIBLE
CPU	on-chip	ARM Cortex-M4F CPU	×1	arm,cortex-m4f
ADC	on-chip	Nordic Semiconductor nRF family SAADC node	×1	nordic,nrf-saadc
ARM architecture	on-chip	Nordic UICR (User Information Configuration Registers)	×1	nordic,nrf-uicr
	on-chip	Nordic EGU (Event Generator Unit)	×6	nordic,nrf-egu
	on-chip	Nordic nRF family ACL (Access Control List)	×1	nordic,nrf-acl
	on-chip	Nordic nRF family MWU (Memory Watch Unit)	×1	nordic,nrf-mwu
Audio	on-chip	Nordic PDM (Pulse Density Modulation interface)	×1	nordic,nrf-pdm
Clock control	on-chip	Nordic nRF clock control node	×1	nordic,nrf-clock
	on-chip	Nordic nRF HFCLK clock control node	×1	nordic,nrf-clock-hfclk
	on-chip	Nordic nRF LFCLK clock control node	×1	nordic,nrf-clock-lfclk
	on-chip	Nordic nRF high-frequency crystal oscillator (nRF52 series)	×1	nordic,nrf52-hfxo
Comparator	on-chip	Nordic nRF COMP (analog COMParator)	×1	nordic,nrf-comp
Counter	on-chip	Nordic nRF timer node	×5	nordic,nrf-timer

TYPE	LOCATION	DESCRIPTION		COMPATIBLE
Cryptographic accelerator	on-chip	Nordic ECB (AES electronic codebook mode encryption)	×1	nordic,nrf-ecb
	on-chip	Nordic nRF family CCM (AES CCM mode encryption)	×1	nordic,nrf-ccm
	on-chip	ARM TrustZone CryptoCell 310	×1	arm,cryptocell-310
Debug	on-chip	ARMv7 instrumentation trace macrocell	×1	arm,armv7m-itm
Display	on-board	Sitronix ST7789V display controller	×1	sitronix,st7789v
Flash controller	on-chip	Nordic NVMC (Non-Volatile Memory Controller)	×1	nordic,nrf52-flash-cont…
	on-chip	Properties defining the interface for the Nordic QSPI peripheral	×1	nordic,nrf-qspi
GNSS	on-board	Generic GNSS NMEA receiver	×1	gnss-nmea-generic
GPIO & Headers	on-chip	NRF5 GPIOTE ×1		nordic,nrf-gpiote
	on-chip	NRF5 GPIO ×2		nordic,nrf-gpio
I2C	on-chip	Nordic nRF family TWIM (TWI master with EasyDMA)	×2	nordic,nrf-twim
I2S	on-chip	Nordic I2S (Inter-IC sound interface)	×1	nordic,nrf-i2s
IEEE 802.15.4	on-chip	Nordic nRF IEEE 802.15.4 node	×1	nordic,nrf-ieee802154
Input	on-board	Group of GPIO-bound input keys	×1	gpio-keys
Interrupt controller	on-chip	ARMv7-M NVIC (Nested Vectored Interrupt Controller)	×1	arm,v7m-nvic
LED	on-board	Group of GPIO-controlled LEDs	×1	gpio-leds
LED strip	on-board	Worldsemi WS2812 LED strip, GPIO binding	×1	worldsemi,ws2812-gpio

TYPE	LOCATION	DESCRIPTION		COMPATIBLE
LoRa	on-board	Semtech SX1262 LoRa Modem	×1	semtech,sx1262
Miscellaneous	on-chip	Nordic FICR (Factory Information Configuration Registers)	×1	nordic,nrf-ficr
	on-chip	Nordic nRF family PPI (Programmable Peripheral Interconnect)	×1	nordic,nrf-ppi
MTD	on-chip	Flash node	×1	soc-nv-flash
Networking	on-chip	Nordic nRF family RADIO peripheral	×1	nordic,nrf-radio
	on-chip	Nordic nRF family NFCT (Near Field Communication Tag)	×1	nordic,nrf-nfct
Pin control	on-chip	Nordic nRF family Pin Controller	×1	nordic,nrf-pinctrl
Power management	on-chip	Nordic nRF power control node	×1	nordic,nrf-power
PWM	on-chip	nRF PWM	×4	nordic,nrf-pwm
	on-chip	nRFx S/W PWM	×1	nordic,nrf-sw-pwm
Regulator	on-chip	Nordic nRF5X regulator (fixed stage of the core supply)	×1	nordic,nrf5x-regulator
	on-chip	Nordic nRF52X regulator (high voltage stage of the main supply)	×1	nordic,nrf52x-regulator…
Retained memory	on-chip	Nordic GPREGRET (General Purpose Register Retention) device	×2	nordic,nrf-gpregret
RNG	on-chip	Nordic nRF family RNG (Random Number Generator)	×1	nordic,nrf-rng
RTC	on-chip	Nordic nRF RTC (Real-Time Counter)	×3	nordic,nrf-rtc

TYPE	LOCATION	DESCRIPTION		COMPATIBLE
Sensors	on-chip	Nordic nRF family TEMP node	×1	nordic,nrf-temp
	on-chip	Nordic nRF quadrature decoder (QDEC) node	×1	nordic,nrf-qdec
Serial controller	on-chip	Nordic nRF family UARTE (UART with EasyDMA)	×1	×1 nordic,nrf-uarte
SPI	on-chip	Nordic nRF family SPIM (SPI controller with EasyDMA)	×1	×2 nordic,nrf-spim
	on-chip	Nordic nRF family SPI (SPI controller)	×1	nordic,nrf-spi
SRAM	on-chip	Generic on-chip SRAM	×1	mmio-sram
Timer	on-chip	ARMv7-M System Tick	×1	arm,armv7m-systick
USB	on-chip	Nordic nRF52 USB device controller	×1	nordic,nrf-usbd
Watchdog	on-chip	Nordic nRF family WDT (Watchdog Timer)	×1	nordic,nrf-wdt

See the Heltec Mesh Node T114 pages [1] for a complete list of board hardware features.

Connections and IOs

LED

- LED4 (green) = P1.03

Push buttons

- BUTTON0 = SW1 = P1.10

System Requirements

- Host machine with USB port
- USB-C cable for programming and serial console (CDC ACM)

- Optional: J-Link or other SWD debugger for debugging
- Optional: UF2 bootloader for drag-and-drop flashing; build with the `heltec_t114_v2/nrf52840/uf2` board target

Programming and Debugging

The `heltec_t114_v2` board supports the runners and associated west commands listed below.

	flash	debug	debugserver	rtt	reset	attach
jlink	✅ (default)	✅ (default)	✅	✅	✅	✅
pyocd	✅	✅	✅	✅	✅	✅

The default `heltec_t114_v2/nrf52840` target uses the standard Nordic nRF52840 partition layout (full flash image, no Nordic nRF5 SDK bootloader / MBR link offset). Use the `heltec_t114_v2/nrf52840/uf2` variant when building for the board's UF2 bootloader and its application partition layout.

Applications for the `heltec_t114_v2/nrf52840` board configuration can be built, flashed, and debugged in the usual way. See Building an Application and Run an Application for more details on building and running.

Flashing

First, run your favorite terminal program to listen for output (e.g. 115200 baud). Then build and flash. Example for the Blinky application:

```
west build -b heltec_t114_v2/nrf52840 samples/basic/blinky
west flash
```

For the UF2 bootloader variant (drag-and-drop firmware):

```
west build -b heltec_t114_v2/nrf52840/uf2 samples/basic/blinky
west flash
```

Debugging

Debugging is supported via SWD. Use an nRF52-compatible probe (e.g. nRF52840 DK, J-Link) and connect SWDIO, SWDCLK, GND, and VDD. Refer to the Nordic nRF5x Segger J-Link page to learn about debugging Nordic boards with a Segger IC. OpenOCD and Zephyr's debug runner can also be used with the appropriate board configuration.

References

[1] *(1, 2)* https://heltec.org/project/mesh-node-t114/